Arcs complete outside a prescribed conic

An exact defect identity and the case of $\mathbb{F}_{16}$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$. We study arcs $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered. The identity includes equality and stability criteria. Write $I_{\mathcal{C}}(A)$ for the total incidence of the secants of A with $\mathcal{C}$. The conic specialization gives

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$-complete k-arc, together with an exact equality criterion and a quantitative stability estimate. In particular, for every prime power q,

$$\rho_{\mathcal{C}}(q) \ge \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

Conversely, projective averaging shows that an ordinary complete b-arc has an image disjoint from a prescribed set H when $b|H| < q^2 + q + 1$; in particular, $\rho_{\mathcal{C}}(q) \le t_2(2,q)$ whenever $t_2(2,q) \le q$. Explicit witnesses and the lower bound give

$$\rho_{\mathcal{C}}(8) = \rho_{\mathcal{C}}(9) = \rho_{\mathcal{C}}(11) = 6, \qquad \rho_{\mathcal{C}}(16) = 9.$$

The last equality follows from a computer-assisted, certificate-checked covering list of 2633 eight-arc representatives over $\mathbb{F}_{16}$: no nonzero quadratic contains an eight-arc's ordinary-uncovered locus while avoiding the arc.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [3], Kim–Vu [12], and Nagy [15].

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is *$\mathcal{C}$-complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

For an arbitrary arc disjoint from $\mathcal{C}$, its uncovered locus is

$$\mathcal{U}_{\mathcal{C}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{C} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum is attained because a maximum-cardinality arc disjoint from $\mathcal{C}$ is maximal under insertion. Equivalently, A is maximal among arcs in the point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The problem differs from two nearby standard notions. A saturating set need not be an arc and has no prescribed exceptional locus. An almost-complete subset of a conic, in the sense of Bartoli–Davydov–Marcugini–Pambianco [5], consists of selected points *on* the conic whose chords cover the complementary points. Ng–Wild's arcs covering a line [16] prescribe a one-dimensional set that must be covered, rather than an exceptional set that may remain uncovered. Here the selected arc is required to avoid the conic.

There is also classical work on *complete exterior sets*: sets of $(q+1)/2$ exterior points whose pairwise joins miss the conic [6, p. 143]. This condition constrains the joins of the selected points, whereas Definition 1.1 requires those joins to cover the complementary off-conic locus. The two conditions therefore point in opposite directions. If

$$U(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right)$$

is the ordinary-uncovered locus, then

$$\begin{aligned} \text{complete exteriority} &\implies \mathcal{C}(\mathbb{F}_q) \subseteq U(A), \\ \mathcal{C}\text{-completeness} &\iff U(A) \subseteq \mathcal{C}(\mathbb{F}_q), \\ \text{both conditions} &\implies U(A) = \mathcal{C}(\mathbb{F}_q). \end{aligned}$$

Strict containment at $q = 7$ and the exceptional equality at $q = 11$ return in Section 8.

The point index with respect to an arc, and the first two double-counting equations for its distribution, are classical; compare Hirschfeld [11, Chapter 9] and Ball [3]. The use of secant counts to bound the smallest complete (k,n)-arcs is also classical; for a recent explicit lower bound see Alabdullah–Hirschfeld [1]. The main observation is that, after splitting those equations over a prescribed exceptional set and its complement, the inequalities normally used to control overlap have an exact remainder. This produces:

(i) an exact prescribed-hole defect identity, with equality and stability;

(ii) conic lower bounds, including the additive asymptotic term $3/2$, and a projective-averaging upper-bound transfer; and

(iii) a certificate-checked quadratic obstruction giving $\rho_{\mathcal{C}}(16) = 9$.

Sections 2–4 establish the defect identity and conic lower bounds. Sections 6, 7, and 8 provide the upper transfer, structural constraints, and the exact value at $q = 16$. Section 5 and the final $q = 11$ subsection are secondary coding applications and may be skipped without affecting the geometric argument.

The classical inputs are the secant-index equations, the Lunelli–Sce $\sqrt{2q}$ scale, the arc–MDS correspondence, and the known $q = 11$ Clebsch geometry. The contributions here are the prescribed-hole remainder and its consequences, the uncovered-locus evaluation obstruction, and the certified relative-conic conclusions stated above.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, and let A be a k-arc in Π. Write $r_A(x)$ for the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor .$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. □

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. □

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered, and define

$$\begin{aligned} V_{\mathcal{H}}(A) &= \Pi \setminus (A \cup \mathcal{H}), \\ \mathcal{X}_{\mathcal{H}}(A) &= \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}, \\ \mathcal{U}_{\mathcal{H}}(A) &= V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A), \\ I_{\mathcal{H}}(A) &= \sum_{y \in \mathcal{H}} r(y). \end{aligned}$$

Thus $\mathcal{X}_{\mathcal{H}}(A)$ is the covered required locus, and $\mathcal{U}_{\mathcal{H}}(A)$ is the uncovered required locus.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity)**.** *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y \in \mathcal{H}} \binom{r(y)}{2}.$$

Substitution and termwise simplification yield

$$m(r-1) - 2\binom{r}{2} = (r-1)(m-r),$$
$$(m-1)r - 2\binom{r}{2} = r(m-r),$$

which proves (5). Every term is nonnegative by Lemma 2.1. □

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \leq N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \geq |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \geq 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). □

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes). *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2+q+1-k-h$. Substitute this equality into (6). $\square$

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability). *Assume $m \ge 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\},$$
$$J = \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}.$$

Then

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). $\square$

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m. This is the precise stability content behind the equality heuristic.

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities). *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \ge q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{10}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{11}$$

The first correction in (11) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Define

$$L_1(q) = \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1)\right\},$$
$$L_2(q) = \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\}.$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \tag{12}$$

The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}(k(q-1)-(k-2)(k-3)), & k \text{ even}, \\ \dfrac{k}{2}((k-1)(q-1)-(k-2)(k-3)), & k \text{ odd}. \end{cases} \tag{13}$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2,16)$ has at least eight points.

Theorem 4.2 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{14}$$

Consequently,

$$\liminf_{q\to\infty}(\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \tag{15}$$

Proof. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (14) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (11) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}(k(q-1)-(k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is exactly

$$\frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2}.$$

Only the leading coefficient matters asymptotically; we bound the remaining coefficients uniformly on $0 \leq a \leq 2$. For $0 \leq a \leq 2$, the last three coefficients are at most $5/2$, 1, and 3, respectively. Since $s \geq 2$,

$$\frac{5}{2}s^2 + s + 3 \leq 4s^2,$$

so their total contribution is at most $4s^2$. The displayed difference is nonnegative, and hence

$$0 \leq \frac{2a-3}{4}s^3 + 4s^2.$$

Dividing by s^2 gives $a \geq 3/2 - 8/s$, which is (14). Letting q tend to infinity gives (15). $\square$

Remark 4.3 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \le I_{\mathcal{C}}(A) \le 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (11) can be useful for finely balanced finite cases, but cannot improve the additive constant 3/2 in Theorem 4.2. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 Coding interpretation

This section is independent of the geometric proof of $\rho_{\mathcal{C}}(16) = 9$ and may be skipped on a first reading. The same remainder has a coding interpretation. Choose nonzero column representatives of a k-arc $A \subset \mathrm{PG}(2, q)$ and form a $3 \times k$ matrix H_A. For $k \ge 4$, the code

$$C_A = \{c \in \mathbb{F}_q^k : H_A c^{\mathsf{T}} = 0\}$$

is a $[k, k-3, 4]_q$ maximum-distance-separable (MDS) code: every three columns are independent, the columns span $\mathbb{F}_q^3$, and any four columns support a dependence. Conversely, a projective parity-check system for a code with these parameters is a k-arc. This correspondence and its coset-weight consequences are classical; see Davydov–Marcugini–Pambianco [8].

We call such a code *projectively GRS* when, after permuting and scaling columns and applying a projective coordinate change, its projective parity-check columns lie on a normal rational curve. In $\mathrm{PG}(2, q)$ this is a nonsingular conic, by the standard normal-rational-curve/GRS correspondence [18]. The next proposition translates an uncovered point into a distance-three syndrome direction and a secant index into a minimum-leader count; this dictionary will be used for the $q = 11$ witness in Section 8.

Proposition 5.1 (Syndrome form of relative completeness)**.** *Let $s \ne 0$ have projective direction $x = [s] \in \mathrm{PG}(2, q)$. Then:*

(i) the minimum weight of a word c with $H_A c^{\mathsf{T}} = s$ is one, two, or three according as $x \in A$, $x \notin A$ lies on a secant of A, or x is ordinarily uncovered;

(ii) if the minimum weight is two, the number of weight-two leaders is exactly $r_A(x)$;

(iii) if the minimum weight is three, the number of weight-three leaders is exactly $\binom{k}{3}$; and

(iv) adjoining x as one more projective parity-check column preserves the MDS property exactly when x is ordinarily uncovered.

Consequently, A is complete outside a prescribed set $\mathcal{H}$ exactly when all its projective distance-three syndrome directions lie in $\mathcal{H}$ (and $A \cap \mathcal{H} = \varnothing$). These directions are deep-hole directions only when covering radius three has separately been established.

Proof. A weight-one representation is a scalar multiple of a selected column. A weight-two representation uses a unique unordered column pair whose projective span is a secant through x; independence of the pair makes its two coefficients unique.

If neither case occurs, any three columns form a basis and represent s. For each three-subset the representation is unique, and none of its coefficients can vanish, so there are exactly $\binom{k}{3}$ weight-three leaders. The extension statement is the same three-column independence test. The leader count also follows as the $d = 4$ case of the general farthest-coset formula in Davydov–Marcugini–Pambianco [8, Theorem 7.7]; the deep-hole/MDS-extension formulation is standard in the Reed–Solomon setting [13]. $\square$

Thus $r_A(x)$ is the exact number of minimum weight-two leaders of every affine syndrome on direction x. All $q-1$ nonzero scalar multiples of a projective syndrome direction have the same distance and leader count: scalar multiplication bijects their coefficient words without changing Hamming support. This accounts for every projective-to-affine factor of $q-1$ below.

The first moment counts leaders, the second counts collisions between distinct leader supports, and Theorem 3.1 is an exact leader-collision remainder after prescribed directions are removed. Thus Corollary 3.3 is also a length obstruction for projective codimension-three MDS codes whose distance-three syndrome directions are confined to the prescribed projective set.

6 An upper-bound transfer from complete arcs

Let $t_2(2,q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2,q)$.

Theorem 6.1 (Projective averaging)**.** *Let $H \subseteq \mathrm{PG}(2,q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{16}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3,q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2+q+1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. $\square$

Corollary 6.2 (Conic transfer)**.** *If $t_2(2,q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2,q). \tag{17}$$

More generally, every complete b-arc with $b \le q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q+1$. For integral $b \le q$ one has $b(q+1) < q^2+q+1$, and Theorem 6.1 applies. $\square$

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [12]. This is less than q for large q, so Corollary 6.2 gives the following immediate consequence.

Corollary 6.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

7 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition 7.1 (Case $\nu \in A$)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ A-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{18}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the tangent A-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (18) follows from Corollary 4.1. □

Proposition 7.2 (Case $\nu \notin A$)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the tangent A-secants are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{19}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of tangent A-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. □

Corollary 7.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (11) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition 7.1 gives $I_{\mathcal{C}}(A) \geq k-1 \geq 1$. Otherwise Proposition 7.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. □

These two cases are useful structural reductions, but Remark 4.3 limits what the incidence term alone can accomplish. For example, at $(q,k) = (16,8)$ inequality (11) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

8 Verified finite-field examples

The verified values at $q = 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 16$, the same ingredients give $8 \le \rho_{\mathcal{C}}(16) \le 9$; a checked projective classification excludes eight.

8.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [10] and Ball [4]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

Lemma 8.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. □

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

The more general problem of avoiding several selected points at once has a sharp finite-field answer, recorded in Appendix B.

8.2 Small values and the classification over $\mathbb{F}_{16}$

Proposition 8.2. *For the standard conic*

$$\mathcal{C} : \; XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(8) = 6, \qquad \rho_{\mathcal{C}}(9) = 6, \qquad \rho_{\mathcal{C}}(11) = 6.$$

Proof. Equation (12) gives

$$L_2(8) = L_2(9) = L_2(11) = 6.$$

Appendix C lists $\mathcal{C}$-complete arcs of size six in each plane. Their verification is described below. □

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

We first state the completeness contract used by the finite certificate. Index the 273 projective points by their unique normalized representatives $[0:0:1]$, $[0:1:z]$, and $[1:y:z]$. The standard four-frame is

$$F = \{[0:0:1], [0:1:0], [1:0:0], [1:1:1]\}.$$

Lemma 8.3 (Augmentation certificate contract)**.** *For $4 \le i \le 8$, let $\mathcal{L}_i$ be a finite list of i-arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $P \in \mathcal{L}_i$, every $x \notin P$ for which $P \cup \{x\}$ is an arc, and every $4 \le i < 8$, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto P'. Then every eight-arc in* $\mathrm{PG}(2,16)$ *is projectively equivalent to a member of $\mathcal{L}_8$.*

Proof. Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to F. Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. □

Theorem 8.4 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in* $\mathrm{PG}(2,16)$*, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \text{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Certified covering lists. Starting from F, the checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. For every listed parent, the checker tests all 273 indexed points. Each legal move carries a child index, an invertible 3×3 matrix, and pointwise nonzero scalar witnesses proving that the induced projectivity maps the enlarged parent onto that child. A separate check matches the certificate books with all parents in the current list. Lemma 8.3 therefore proves exhaustiveness without trusting the generator's canonical labels or deduplication. As an external consistency check, the final list length agrees with the projective-class count of Al-Seraji–Al-Ogali [2, Theorem 3.8.1].

Leaf obstruction. For 2630 listed leaves, the certificate gives six ordinarily uncovered points and an explicit inverse to their 6×6 quadratic evaluation matrix. This is the first alternative of Lemma 8.1. For each remaining leaf, a checked linear combination expresses evaluation at a selected arc point in the span of evaluations at ordinarily uncovered points, giving the second alternative. Thus every listed leaf has the asserted quadratic-avoidance property.

Projective transport. If gA is a listed representative, replace Q by $Q \circ g^{-1}$. This form is nonzero, its zero set is $gZ(Q)$, and projectivities carry secants to secants, so $U(gA) = gU(A)$. A quadratic as in the statement would therefore contradict the checked obstruction for the listed leaf. Appendix A records the independent regeneration and formal trust boundary. □

Corollary 8.5 (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq$* $\mathrm{PG}(2,16)$*,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (12) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 8.4. The nine-point witness in Appendix C gives the matching upper bound. □

Remark 8.6 (Scope of the classification certificate). The proof requires an exhaustive covering list, not pairwise inequivalence of its entries. Pairwise deduplication and the exact quotient-class count belong to the cited ordinary classification; their agreement with the 2633-entry covering list is a consistency check, not part of the trusted exhaustiveness argument.

8.3 A secondary coding application over $\mathbb{F}_{11}$

This subsection is independent of the proof of $\rho_{\mathcal{C}}(16) = 9$. The six-point witness over $\mathbb{F}_{11}$ is the classical Clebsch hexagon with icosahedral A_5 symmetry; see Edge, Dye, and Storme–Van Maldeghem for the configuration and its orbit geometry [7, 9, 17]. Dye's edge criterion and the complete-exterior formulation of Blokhuis–Seress–Wilbrink show that its secants avoid the conic, so $\mathcal{C}(\mathbb{F}_{11}) \subseteq U(A)$ [9, discussion preceding Theorem 6, p. 281] [6, pp. 143, 146]. The verified relative completeness of the witness gives the reverse inclusion and hence $U(A) = \mathcal{C}(\mathbb{F}_{11})$.

This equality is exceptional. For comparison, every four-arc has three repeated off-arc intersections among its six secants, and therefore

$$|U(A)| = (q^2 + q + 1 - 4) - (6(q - 1) - 3) = (q - 2)(q - 3).$$

The exterior four-arc over $\mathbb{F}_7$ has $|U(A)| = 20 > 8 = |\mathcal{C}(\mathbb{F}_7)|$, so its conic lies strictly inside the uncovered locus [6, pp. 143, 146].

The verified incidence statistics place the $q = 11$ equality among the other exact examples:

q	$\lvert A\rvert$	$\binom{\lvert A\rvert}{2}$	$q^2 - \lvert A\rvert$	$I_{\mathcal{C}}(A)$	$\max_{x\notin A} r(x)$
8	6	15	58	16	3
9	6	15	75	13	3
11	6	15	115	0	3
16	9	36	247	32	4

The row at $q = 11$ has a stronger geometric interpretation. Since $I_{\mathcal{C}}(A) = 0$, none of the 15 secants meets $\mathcal{C}$: every secant of the witness is exterior to the conic. If N_i denotes the number of required points of index i, relative completeness and the maximum-index bound give only $i = 1, 2, 3$, while the two moment equations give

$$N_1 + N_2 + N_3 = 115, \qquad N_1 + 2N_2 + 3N_3 = 150, \qquad N_2 + 3N_3 = 45.$$

Consequently $(N_1, N_2, N_3) = (90, 15, 10)$.

Under the dictionary of Section 5, the equality $U(A) = \mathcal{C}(\mathbb{F}_{11})$ says that the associated code has precisely the conic as its projective distance-three syndrome locus. The next proposition records the resulting code and coset data.

Proposition 8.7 (The $q = 11$ code and coset spectrum). *Let A be the six-point witness in Appendix C, and let C_A be its parity-check code. Then:*

(i) C_A is a projectively non-GRS $[6, 3, 4]_{11}$ MDS code of covering radius three, and its projective deep-hole syndrome locus is exactly the twelve-point conic $\mathcal{C}$;

(ii) the affine cosets of distances $0, 1, 2, 3$ *number* $(1, 60, 1150, 120)$*, and the distance-two cosets split as* $(900, 150, 100)$ *according as they have one, two, or three minimum-weight leaders.*

Proof. Every witness triple has nonzero determinant and the six columns span $\mathbb{F}_{11}^3$, giving the MDS parameters. The 6×6 evaluation matrix of the quadratic monomials

$$X^2,\ XY,\ XZ,\ Y^2,\ YZ,\ Z^2$$

is nonsingular, so no conic contains all six projective columns; by the preceding definition and the classical normal-rational-curve dictionary, the code is not projectively GRS. Direct finite arithmetic gives the secant-index spectrum

$$0 : 12, \qquad 1 : 90, \qquad 2 : 15, \qquad 3 : 10,$$

where index zero is precisely the conic. Multiplying each nonzero projective direction by its ten affine scalars gives all nonzero affine syndromes bijectively. Proposition 5.1 identifies the determinant-zero witness pairs with weight-two leader supports; uniqueness on an independent support and scalar multiplication then give the claimed coset and leader counts.

The twelve index-zero directions yield affine syndromes of distance three. Every syndrome has distance at most three because any three parity-check columns form a basis. Hence the covering radius is three, and these twelve directions are exactly the projective deep-hole locus. □

Optional extension structure. Relative completeness confines every further projective MDS extension to the twelve conic points. On those points, two extensions conflict exactly when their chord contains a witness point; the residual extension problem is therefore a finite graph-independence problem.

Proposition 8.8 (The $q = 11$ extension complex)**.** *For the same witness:*

(i) the simultaneous one-column MDS extensions form the independence complex of the icosahedral graph, with independence polynomial

$$1 + 12t + 36t^2 + 20t^3;$$

exactly six two-column and twenty three-column extensions are maximal, giving six complete eight-arcs and twenty complete nine-arcs containing the fixed seed, and there is no ten-arc extension;

(ii) each witness column colours five pairwise disjoint edges of the icosahedron, missing an antipodal pair, and the six colour classes partition its thirty edges. Adding the six antipodal edges turns them into six perfect matchings of the augmented graph.

Proof. For simultaneous extensions, two conic columns conflict exactly when their chord contains a witness column. No three conic columns are collinear, so there is no additional triple obstruction and valid extension sets are exactly graph-independent sets. Exhausting the 2^{12} subsets gives the displayed polynomial and maximal-set counts. The same determinant table partitions the thirty conflict edges into the six stated matchings. Thus maximality in the independence complex upgrades to ordinary completeness. The six missing antipodal edges are distinct; adjoining the appropriate one to each colour class gives six disjoint perfect matchings of the augmented graph. Appendix A records the two independent implementations and the formal checks. □

9 Further questions

The results leave four concrete directions.

Problem 9.1. Determine whether the uncovered-quadratic-rank obstruction used at $q = 16$ extends to other even orders or gives useful lower bounds beyond the scalar defect inequality.

Problem 9.2. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 6.1 currently supplies only the general complete-arc bound $O(\sqrt{q}(\log q)^c)$.

Problem 9.3. Classify equality and bounded-defect configurations in Theorem 3.1, and determine whether the conic stabilizer yields finite-order restrictions not reducible to a bound on $I_{\mathcal{C}}(A)$. Remark 4.3 shows that such a bound alone cannot improve the asymptotic constant obtained here.

Problem 9.4. Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\mathrm{PG}(n, q)$. The evaluation lemma already applies to arbitrary feature maps and Veronese degrees; the missing ingredient is an appropriate exact secant-moment remainder for caps.

The central distinction is between universal overlap, already measured by the classical second index equation, and genuinely conic-specific geometry. The former gives the lower bound in this paper. Progress beyond it will probably require an explicit construction, an algebraic obstruction, or a classification of near-extremal secant arrangements rather than another application of the same incidence budget.

A Computational and formal verification

The body separates mathematical reductions from finite exclusion. The corresponding evidence and trust boundaries are summarized below. The source supplement's `TRUST.md` manifest records exact commands, toolchain versions, SHA-256 digests, theorem names, and frozen outputs.

Claim	Evidence	Trust boundary
Four upper-bound witnesses	Exact Python verifier checks the field model, conic, arc condition, coverage, and both secant moments	Certificate check of the displayed coordinates; no search-completeness claim
General theorem chain	Companion Lean library formalizes the incidence, defect, conic, asymptotic, averaging, nucleus, evaluation, syndrome, and coding arguments	Kernel checking with axioms `propext`, `Classical.choice`, and `Quot.sound`; no `sorry`, custom axiom, or `native_decide`
$q = 16$ classification	C++ generator emits local transition and quadratic-evaluation data for covering lists of lengths 4/61/454/2633	Lean checks transition coverage, leaf rejection, frame normalization, and projective transport; generator assertions, canonical labels, and deduplication are not trusted
$q = 11$ code and extension data	Lean checks the stated code, syndrome, and graph claims; independently written Python and C++ programs regenerate the finite data and include coordinate-invariance and mutation controls	The propositions rest on kernel-checked statements; the two programs are independent reproducibility checks

The witness entry point is `verify_relative_conic_arcs.py`. The classification entry point is `search_rhoc16.cpp`; its checked-in report and emitted Lean data fix the finite search input. For the upper-bound witnesses, a separate rules-only Lean checker verifies disjointness, the arc condition, and coverage of the canonical projective representatives $[1 : y : z]$, $[0 : 1 : z]$, and $[0 : 0 : 1]$, and a soundness theorem transports acceptance to relative completeness. The finite examples do not depend on the external Kim–Vu hypothesis used only for the conditional asymptotic upper bound.

The three exceptional leaves. Using the binary $\mathbb{F}_{16}$ encoding from Appendix C and coefficient order $(X^2, Y^2, Z^2, XY, XZ, YZ)$, the one-dimensional kernels of the three exceptional evaluation matrices are generated by

$$(1, 1, 1, 1, 1, 0), \qquad (0, 0, 0, 5, 4, 1), \qquad (2, 1, 1, 5, 5, 1).$$

The first and third forms split over $\mathbb{F}_{16}$ as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z),$$

and each union of lines contains two selected points. The middle form is nonsingular and contains seven selected points. Thus two rank defects come from reducible quadratics, while the third comes from an arc differing by one point from a seven-point conic subset. These descriptions are not used for classification completeness.

B A general finite-field evaluation dichotomy

The one-selected-point obstruction used in Section 8 has a sharp extension to several selected points. The vector-space covering threshold is classical; see, for example, Khare [14].

Proposition B.1 (Finite-field evaluation dichotomy)**.** *Let W be a finite-dimensional vector space over $\mathbb{F}_q$, let $\nu : X \to W$ be any feature map, and let $U, A \subseteq X$ be finite with $|A| \leq q$. There is a linear form $f \in W^*$ such that*

$$f(\nu(u)) = 0 \quad (u \in U), \qquad f \neq 0, \qquad f(\nu(a)) \neq 0 \quad (a \in A)$$

if and only if

$$\langle \nu(U) \rangle \neq W \quad \text{and} \quad \nu(a) \notin \langle \nu(U) \rangle \quad (a \in A).$$

The uniform threshold is sharp: $q+1$ distinct hyperplanes through a common codimension-two subspace can cover W.

Proof. Put $S = \langle \nu(U) \rangle$. Necessity follows by applying f to the stated span relations. Conversely, $S \neq W$ makes the annihilator $S^\perp \subseteq W^*$ nonzero. For each $a \in A$, the condition $\nu(a) \notin S$ says that evaluation at $\nu(a)$ cuts out a proper hyperplane of $S^\perp$. At most q proper hyperplanes cannot cover a nonzero vector space over $\mathbb{F}_q$: their union has size at most $1 + q(|S^\perp|/q - 1) < |S^\perp|$. Any form outside their union has the required properties. For sharpness, quotient by a common codimension-two subspace and use all $q+1$ lines in $\mathbb{F}_q^2$. $\square$

For the degree-d Veronese map, the proposition gives the same dichotomy for homogeneous degree-d forms.

C Witness coordinates and field encodings

All coordinates are homogeneous and use the standard conic $XZ = Y^2$.

For $q = 8$, write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ and encode $a_0 + a_1\alpha + a_2\alpha^2$ by the binary integer $a_0 + 2a_1 + 4a_2$. A witness is

$$(0,1,1), (0,1,2), (1,0,1),$$
$$(1,0,2), (1,1,2), (1,1,6).$$

For $q = 9$, write $\mathbb{F}_9 = \mathbb{F}_3[\alpha]/(\alpha^2 + 1)$ and encode $a_0 + a_1\alpha$ by $a_0 + 3a_1$. A witness is

$$(1,0,4), (1,0,5), (1,1,0),$$
$$(1,1,2), (1,2,3), (1,2,4).$$

For $q = 11$, coordinates lie in the prime field. A witness is

$$(1,10,0), (1,9,1), (1,4,7),$$
$$(1,8,5), (0,1,4), (1,1,7).$$

For $q = 16$, write $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and use binary coefficient encoding. A witness of size nine is

$$(1,5,14), (1,5,3), (1,9,2), (1,0,9), (1,13,13),$$
$$(1,6,10), (1,3,9), (0,1,11), (1,10,2).$$

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